



Department of Electrical Electronics Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E/VII/EEE

Course Code & Title: EE8703/Renewable Energy Systems

Name of the Faculty member (s): Mrs G.Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit I / Environmental consequence of Fossil Fuel usage
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** 1a
- **Activity Chosen:** One Minute Paper
- **Justification:**

The basics of conventional and renewable energy sources are discussed. After teaching this topic, I thought of this activity for making to have clarity about the difference between conventional and non conventional energy systems. I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

- **Details of the Implementation:**

After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.

Total Strength :60

Reporter: Myself

At the end of the Class (Last 5 minutes)

I asked the students to think about Environmental consequence of Fossil Fuel usage concept for 3 minutes.

Then I told them to write as much as they can within a short period of time (1 minute)

Finally, I collected the papers from each column

- **CO – PO / PSO mapping:**

CO	PO1	PO6	PO7	PO8	PO9	PO10	PO12
CO1	2	3	3	1	3	3	2

(1 – Low

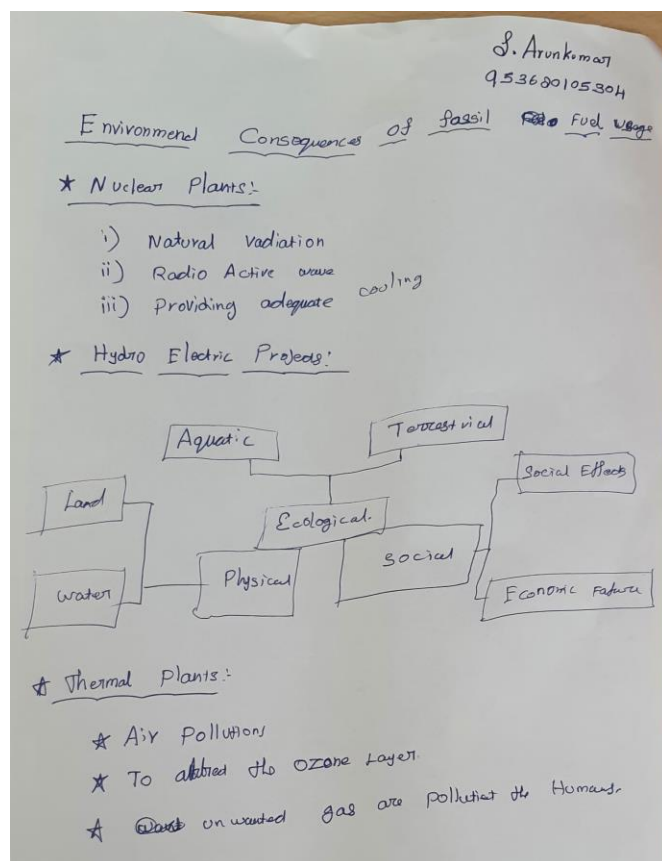
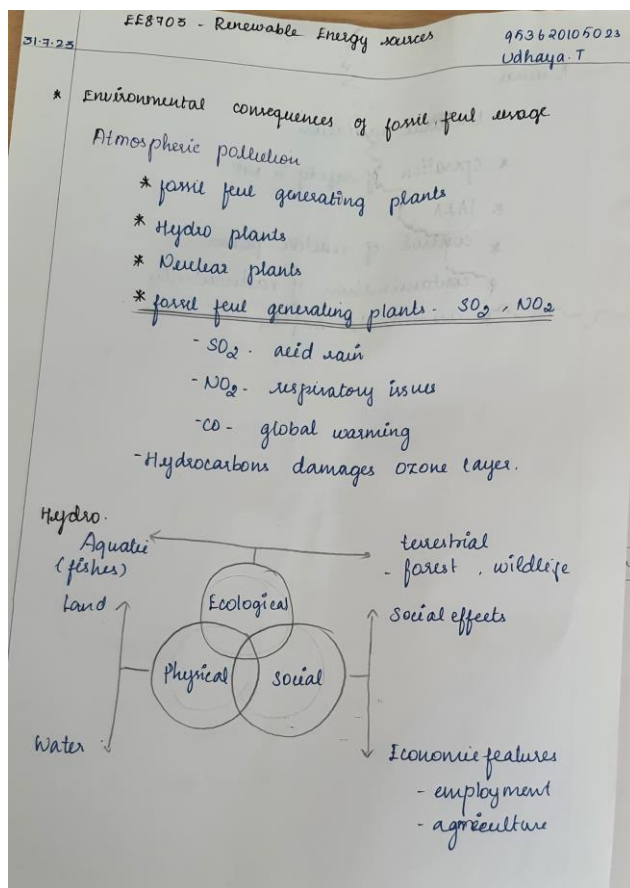
2 – Moderate

3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO6	PO7	PO8	PO9	PO10	PO12
	2	3	3	1	3	3	2
Justification for correlation	The basics of conventional and renewable energy sources are discussed for the engineering problems.	As role of engineering in protecting the globe is discussed	The impact of renewable energy technology in environmental sustainability will be taught to the students	Consequences of unethical environmental practice & method to overcome the same is briefly explained	The students will be given group assignment / presentation	The students will be given group assignment / presentation. They will prepare a report on the same	For preparing the assignment, students need to comprehend few technical contents

• Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to explain Environmental consequence of Fossil Fuel usage.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 90% of students answered.

- ***Benefit of the practice:***

- Students were able to attend the question even in the questions are in indirect form.
- Students were able to explain the concepts in examination without any confusion.

- ❖ ***Challenges faced in implementation:***

- In the class mostly few students hesitate to answer the questions.
- Time utilization for conducting activity.

References:

1. Joshua Earnest, Tore Wizeliu, ‘Wind Power Plants and Project Development’, PHI Learning Pvt.Ltd, New Delhi, 2011.
2. D.P.Kothari, K.C Singal, RakeshRanjan “Renewable Energy Sources and Emerging Technologies”, PHI Learning Pvt.Ltd, New Delhi, 2013.

Signature of Faculty Member

HOD



Department of Electrical Electronics Engineering

Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.E/VII/EEE

Course Code & Title: EE8703/Renewable Energy Systems

Name of the Faculty member (s): Mrs G.Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit II /Wind Energy
- **Course Outcome:** CO2
- **Topic Learning Outcome:** 2a
- **Activity Chosen:** Ungraded Quiz
- **Justification:**

I explained the wind generation principle, types of wind energy systems and the components of Wind Power Plant. After teaching this topic ,I thought of this activity for making to have clarity about the wind Energy. So I can judge the level of understanding.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
 - Total Strength :33
 - Reporter: Myself
 - At the end of the Class (Last 10 minutes)
 - I asked the students to think about the wind generation principle, types of wind energy systems and the components of Wind Power Plant for 2 minutes.
 - Then I told them to write to attend the quiz which is ungraded(8 minutes)
 - Finally, I collected the responses from each students.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2
CO2	2	1	3	2	1	3	3	2	2	1

(1 – Low

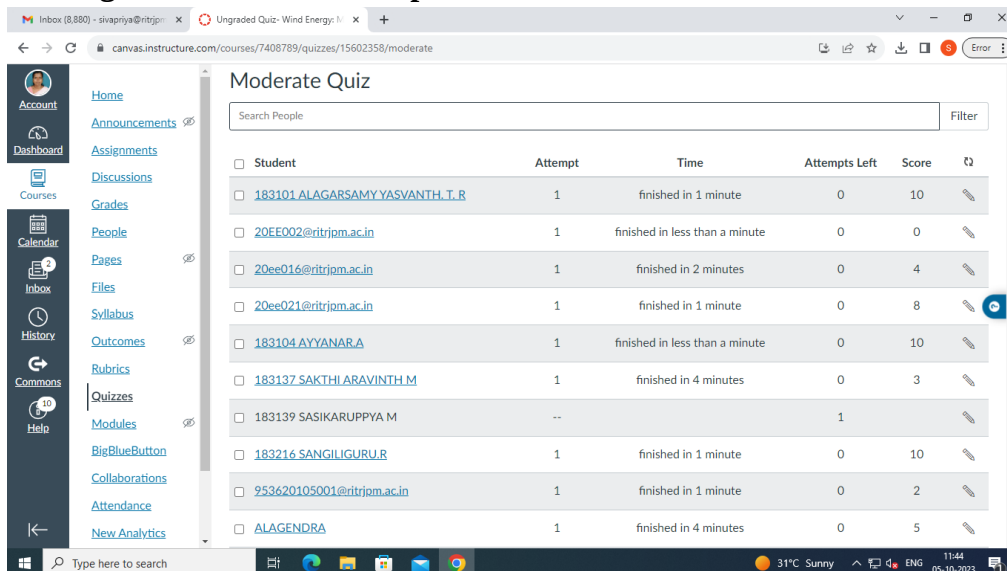
2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO5	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2
	2	1	3	2	1	3	3	2	2	1
Justification for correlation	Students will be solving basic problem on wind energy using basic mechanical concepts	Suitable mathematical and engineering knowledge will be applied by the students to solve basic problems	Students will be using 3M Wind Energy virtual laboratory and gains proficiency on the same	The basic role of wind energy power generation in environmental sustainability is taught to the students	Students follow basic ethical practice to prepare the assignment	Presentation / Assignment will be submitted by the students on wind power plants	Assignments will ensure the submission of a technical report with structured flow	Through assignment, students will explain the minimal and basic change in trends while designing wind farms for given location.	Students will analyze the performance of wind energy conversion systems by using Matlab software	Students will design a wind Power plant system for the given location by analyzing various implementation parameters

• **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

I preferred this activity because it will help me to evaluate what types of questions I am asking students and what level they are working at.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students can understand their knowledge in that particular topic .

- ❖ ***Challenges faced in implementation:***

- Since it is conducted online ,it is very tough to make all students to attend.
 - It is difficult to access whether student have attended the quiz properly or not.

References:

- ❖ https://www.nactateachers.org/images/The_Student-Developed_Quiz_or_Exam-__Scaffolding_Higher-Order_Thinking.pdf
- ❖ Joshua Earnest, Tore Wizeliu, ‘Wind Power Plants and Project Development’, PHI Learning Pvt.Ltd, New Delhi, 2011.
- ❖ D.P.Kothari, K.C Singal, RakeshRanjan “Renewable Energy Sources and Emerging Technologies”, PHI Learning Pvt.Ltd, New Delhi, 2013.

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE

Course Code & Title: EE8703 Renewable Energy Systems

Name of the Faculty member (s): Mrs.G.Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit III / Solar Photovoltaic Energy Conversion, Solar Thermal Collector, Evacuated Type Flat plate collectors
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** 3a,3d
- **Activity Chosen:** Field Visit, Demonstration
- **Justification:**

Solar Photovoltaic (PV) is a technology that converts sunlight (solar radiation) into direct current electricity by using semiconductors. When the sun hits the semiconductor within the PV cell, electrons are freed and form an electric current. Solar PV technology is generally employed on a panel (hence solar panels). After teaching the constructional, I thought of conducting this as field visit for making the students to give overview of PV panel which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 60,

Photographer: one student – Mr. S. Muthuhareesh (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about solar PV system operation and its position for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the operation of solar photovoltaic energy conversion system for another 1 minute.
- ✓ Finally, I randomly asked students about the operation of solar photovoltaic energy conversion system. (2 minutes).



- CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 7	PO 8	PO 9	PO 10	PO12	PSO1	PSO3
CO 3	2	1	2	1	3	3	2	2	1

(1 – Low

2 – Moderate

3 – High)

- PO / PSO mapped:

Innovative practice	PO 7
	2
Justification for correlation	The basic applications of solar energy in an environmental context are taught to the students, hence it is moderately correlated (level 2).

- Images / Screenshot of the practice:





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were satisfied with the field visit demonstration class on Solar PV system, also they said, it increases our curiosity to explore more about the topic.

❖ *Benefit of the practice:*

1. Students are able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked during discussion session of field visit.
3. The success of the activity was evaluated by asking the same question during IAT-II – **Around 80% of students answered.**
4. Students are able to explain the concepts in examination without any confusion.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity, slow learners were not able to answer the question.

References:

1. D.P.Kothari, K.C Singal, RakeshRanjan “Renewable Energy Sources and Emerging Technologies”, PHI Learning Pvt.Ltd, New Delhi, 2013
2. Scott Grinnell, “Renewable Energy & Sustainable Design”, CENGAGE Learning, USA, 2016

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